

CLINICAL PROBLEM-SOLVING

Caren G. Solomon, M.D., M.P.H., *Editor*

Hard to Swallow

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Amy L. Miller, M.D., Ph.D., and Joseph Loscalzo, M.D., Ph.D.*In this Journal feature, information about a real patient is presented in stages (boldface type) to an expert clinician, who responds to the information by sharing relevant background and reasoning with the reader (regular type). The authors' commentary follows.*

A previously healthy 20-year-old college student presented to her pediatrician in September 2019 with a 3-day history of throat pain, malaise, cough, and chills. She had no congestion or rhinorrhea. Her temperature was 37.2°C; vital signs were otherwise unremarkable. Physical examination revealed tender cervical lymphadenopathy, pharyngeal erythema, and tonsillar enlargement that was greater on the left than the right, but there were no exudates. The result of a rapid test for streptococcal antigen was negative. A pharyngeal swab specimen was sent for culture, and the patient was advised to drink fluids and take ibuprofen as needed.

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Patients with viral pharyngitis (the most likely cause of the symptoms) typically present with low-grade fevers, cough, congestion, and coryza. Although infection with severe acute respiratory syndrome coronavirus 2 was not a consideration at the time, it would now be on the differential diagnosis. The Modified (McIsaac) Centor Score can be used to determine whether testing for group A streptococcus is warranted. Testing is pursued if two or more of the following findings are present: tonsillar exudates or enlargement, tender cervical lymphadenopathy, fever, and absence of cough, although a determination of fewer than two symptoms does not exclude group A streptococcus pharyngitis. Since this patient has tonsillar enlargement with cervical lymphadenopathy, testing is warranted.

She returned the following day with worsening throat pain and dysphagia. Her voice sounded muffled, but she was able to speak comfortably, without stridor or respiratory compromise. She was afebrile, and vital signs were within normal limits. Physical examination revealed tonsillar enlargement with ongoing tender cervical lymphadenopathy. The patient lived in a dormitory, and her roommate had had a recent upper respiratory infection. The patient reported no sexual activity and no use of tobacco, alcohol, or illicit substances. She had recently taken nitrofurantoin for a urinary tract infection but was not taking any long-term medications. Her history was notable for attention deficit disorder and methicillin-resistant *Staphylococcus aureus* (MRSA) cellulitis. As a child, she had undergone a left carotid body resection for a benign tumor. She was fully vaccinated with standard routine vaccinations. There was no family history of chronic infections or cancer.

Despite the negative result on the streptococcal antigen test, infection with group A streptococcus remains on the differential diagnosis, since antigen testing has a lower sensitivity than throat culture. The presence of a muffled “hot potato” voice, while difficult to differentiate from dysphonia that is due to laryngitis, suggests

upper-airway compromise. Other red flags in patients with signs of upper-airway compromise that could become life-threatening include hoarseness, drooling, stridor, and respiratory distress. In this patient, asymmetric tonsillar hypertrophy and airway obstruction suggest peritonsillar abscess or a deep-neck-space infection. I would obtain a complete blood count and conduct a heterophile antibody test for mononucleosis consequent to infection with Epstein–Barr virus (EBV).

A heterophile antibody test was nonreactive. A single dose of 10 mg of oral dexamethasone was prescribed, with acetaminophen recommended as needed, given the suspicion of infectious mononucleosis.

Empirical use of glucocorticoids has unclear efficacy, even in patients with documented infectious mononucleosis, and the negative result on the heterophile antibody test makes it unlikely that this patient has an acute EBV infection. I reserve glucocorticoids for hospitalized patients with severe airway obstruction.

The following day, a negative result was returned on the β -hemolytic streptococcal culture. Serologic testing for EBV early antigen was negative. The results of tests for EBV-specific capsid antigens IgM and IgG were positive, as was the result of a test for EBV nuclear antigen.

The negative throat culture makes infection with group A streptococcus unlikely. The serologic analyses suggest the convalescent phase of infectious mononucleosis. EBV nuclear antigen is highly specific for resolved EBV infection and becomes detectable within 4 to 6 weeks after the initial infection. Although tests for EBV capsid antigen IgM are typically positive during the acute phase of mononucleosis, results can also be positive for a wide range of other infectious and inflammatory conditions.

An upper-airway bacterial infection, such as peritonsillar abscess, epiglottitis, or retropharyngeal abscess, seems increasingly likely. Blood cultures and a complete blood count should be obtained, and contrast-enhanced computed tomography (CT) of the neck should be performed. An otolaryngologist should be consulted, and empirical treatment with antibiotic agents should be considered.

Five days later, the patient presented to the emergency department at another hospital with worsening throat pain, a muffled voice, odynophagia, and shortness of breath. She reported no excessive drooling. The heart rate was 122 beats per minute, the blood pressure 107/60 mm Hg, the temperature 37.3°C, the respiratory rate 18 breaths per minute, and the oxygen saturation 99% while she was breathing ambient air. She was not in any obvious distress. Mucous membranes were moist. The tonsils were enlarged, with scant secretion of purulent exudate. She also had diffuse, painful cervical lymphadenopathy, but there was no limitation in neck movement. The pulmonary and abdominal examinations were unremarkable. Cardiovascular examination was notable for tachycardia, but there were no murmurs, rubs, or gallops. There was no rash.

One liter of intravenous fluid was infused, and the patient's heart rate decreased to 106 beats per minute. The white-cell count was 10,400 per cubic millimeter, with 800 atypical lymphocytes per cubic millimeter; the hemoglobin level was 12.2 g per deciliter, and the platelet count 140,000 per cubic millimeter. A basic metabolic panel was normal. No blood cultures were sent. Benzocaine spray was administered for sore throat, and an additional liter of intravenous fluid was provided. She was discharged home once she was able to swallow liquids.

The progressive symptoms and the findings on physical examination suggest a progressive bacterial infection in the airway. Discharge without further evaluation is ill-advised. The mild thrombocytopenia may be an early sign of sepsis. Continued conservative management is likely to result in worsening symptoms, which could be life-threatening.

Two days later, the patient returned to the emergency department. She was unable to clear secretions and noted a worsening cough and throat pain, foul-smelling breath, and pain on the left side of her neck. She also reported three episodes of nonbilious, nonbloody emesis. The patient was anxious and toxic appearing. The temperature was 39.4°C, the heart rate 134 beats per minute, the blood pressure 108/51 mm Hg, the respiratory rate 26 breaths per minute, and the oxygen saturation 99% while the patient was breathing ambient air. Tender cervical lymphadenopathy was present,

and she now had pain on the left side of her neck with rightward neck rotation. Posterior pharyngeal enlargement, without exudates, was noted.

Tachypnea and inability to manage secretions suggest obstruction of the upper airway. Without urgent intervention, airway compromise may progress, and intubation or perhaps emergency tracheostomy may be required. I would obtain a complete blood count, including a differential count, cultures, and arterial blood gas levels. Broad-spectrum antibiotics should be initiated. I would start with vancomycin, with either meropenem or cefepime plus metronidazole. I would also prescribe a single dose of a high-potency glucocorticoid to suppress airway edema. The otolaryngology department should be consulted immediately for endoscopic evaluation, and I would obtain a contrast-enhanced CT scan of the neck as well as a chest radiograph.

The white-cell count was 13,800 per cubic millimeter, with 95.7% neutrophils and no atypical lymphocytes. The hemoglobin level was 11.8 g per deciliter, and the platelet count 27,000 per cubic millimeter. A peripheral-blood smear showed no blasts or schistocytes. The potassium level was 3.0 mmol per liter, bicarbonate 20 mmol per liter, blood urea nitrogen 24 mg per deciliter (8.6 mmol per liter), and creatinine 1.2 mg per deciliter (106 μ mol per liter), up from 1 mg per deciliter (88 μ mol per liter) 2 days earlier. The anion gap was 15. A urine test for human chorionic gonadotropin was negative.

Although there are no schistocytes, which would suggest a microangiopathic hemolytic anemia, I would obtain a fibrinogen level and international normalized ratio (INR) to determine whether the patient is entering a consumptive coagulopathic state. The rising creatinine level, the rising level of blood urea nitrogen, and the metabolic acidosis suggest that she is at risk for septic shock.

A 1000-mg dose of acetaminophen was administered orally. During the next 2 hours, the patient became increasingly agitated. The temperature increased to 41.1°C and the heart rate to 160 beats per minute, and the blood pressure dropped to 86/40 mm Hg. The oxygen saturation while she was breathing ambient air was 97%. A liter of in-

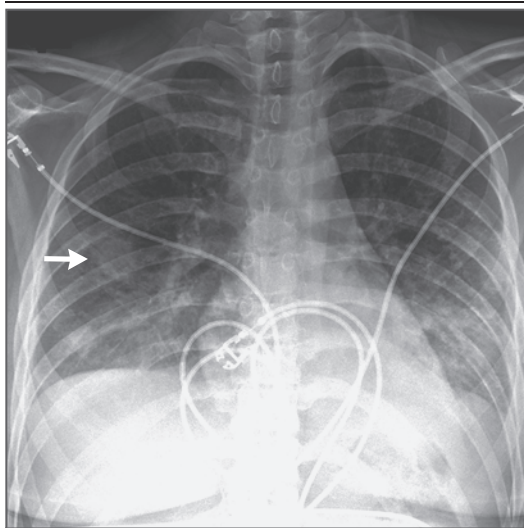


Figure 1. Chest Radiograph.

A chest radiograph shows diffuse hazy opacifications with consolidative opacities in the lower lung fields and a nodular opacity in the right midlung field (arrow).

travenous fluid was administered, and the blood pressure briefly improved to 114/60 mm Hg but then dropped to 98/40 mm Hg. An additional 2 liters of intravenous fluid were administered. Blood cultures were drawn, followed by the administration of 2 g of intravenous ceftriaxone. The lactate level was 4.9 mmol per liter (normal range, 0.5 to 2.0 mmol per liter), the INR 1.2, and the fibrinogen level 573 mg per deciliter (normal range, 116 to 462 mg per deciliter).

The patient's decompensation demonstrates the hazards of delaying diagnostic evaluation and treatment in a potentially life-threatening situation. She is acutely hypotensive, has a dangerously high fever, and shows signs of systemic malperfusion. I am concerned that ceftriaxone is insufficient.

A frontal image of the chest was obtained on portable radiograph (Fig. 1) and showed diffuse, hazy opacifications, consolidative opacities in the lower lung fields, and a nodular opacity in the right midlung field. Contrast-enhanced CT of the neck showed a peritonsillar phlegmon (Fig. 2A) and an occlusive thrombus within the left common facial vein (Fig. 2B). Piperacillin–tazobactam was administered at a dose of 3.375 g, and intravenous methylprednisolone was started at a dose of 60 g. The patient was transferred to this hospital.



The patient has suppurative thrombophlebitis of the internal jugular vein (Lemierre's syndrome). Variants of Lemierre's syndrome can affect any of the large venous structures of the face, and

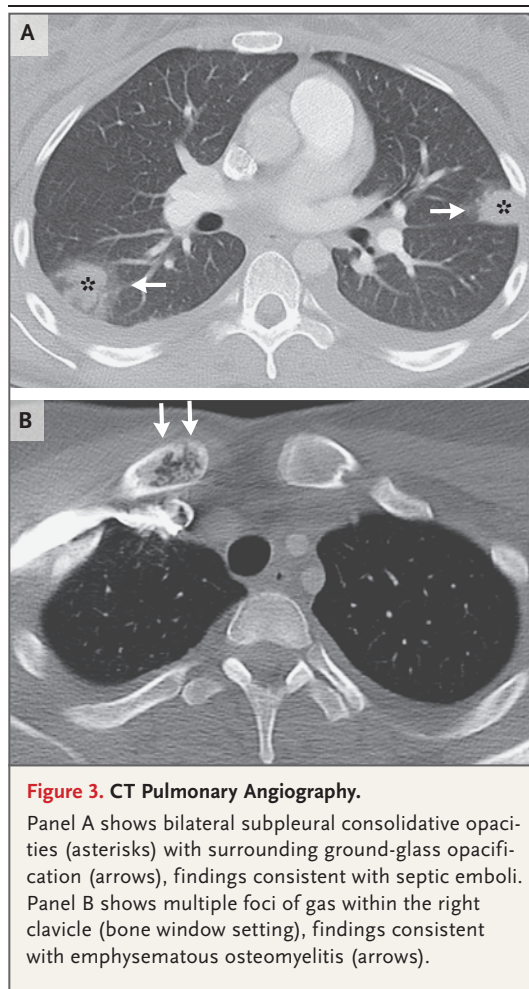
bacterial spread can affect the surrounding facial structures. Antecedent infectious mononucleosis, bacterial pharyngitis, and dental infections are well-documented triggers for this syndrome, in which patients typically present with high-grade fever, neck pain, and respiratory distress. Common sequelae include septic thromboembolism, infiltrates, and infiltrates with cavitation. The nodular opacity suggests a septic embolus and should be evaluated with CT of the chest. Other potential complications include metastatic bacterial dissemination resulting in lung abscess with empyema formation, septic arthritis, or osteomyelitis.

The most common causal bacterium is the gram-negative oral anaerobe *Fusobacterium necrophorum*. Other causal bacteria include *Klebsiella pneumoniae*, *Pseudomonas aeruginosa*, MRSA, *F. nucleatum*, and beta-hemolytic streptococci. Infections can be polymicrobial. Empirical therapy for Lemierre's syndrome should be broad. Given the presence of septic shock, I would recommend starting vancomycin, cefepime, and metronidazole.

The patient was transferred to the intensive care unit. At that time, the temperature was 37.6°C, the heart rate 124 beats per minute, the blood pressure 102/51 mm Hg, the respiratory rate 22 breaths per minute, and the oxygen saturation 94% with the delivery of 10 liters of oxygen by face mask. She was breathing comfortably, and faint bibasilar crackles were present. The results on examination were otherwise unchanged. The venous blood gas was notable for a pH of 7.42. The partial pressure of carbon dioxide was 30 mm Hg, and the partial pressure of oxygen was 49 mm Hg. Vancomycin, cefepime, and metronidazole were administered.

The combination of hypocapnia with hypoxemia indicates acute hypoxemic respiratory failure with adequate alveolar ventilation. Serial blood carbon dioxide levels should be obtained to monitor for hypercarbia, a harbinger of impending upper-airway collapse. I would favor a 24-hour course of dexamethasone to reduce airway edema and would verify the airway patency through visual assessment of the vocal cords on endoscopy.

Nasopharyngeal endoscopy revealed a fullness of the palatine tonsils that was greater in the left tonsil than in the right, a nonedematous epiglottis,



and symmetric vocal cords. There were no mass lesions. Dexamethasone was administered at a dose of 10 mg every 8 hours for the next 24 hours. CT angiography revealed multiple, subpleural, consolidative lesions (one with central cavitation) in both lungs, findings consistent with septic emboli (Fig. 3A). There were also multiple small foci of air within the right clavicle (Fig. 3B).

The clavicular findings, which suggest acute hematogenous emphysematous osteomyelitis, are rare; the air is produced by gas-forming organisms in the bone marrow space. Given the thrombocytopenia, I would not consider anticoagulation.

On hospital day 2, anaerobic cultures grew gram-negative bacilli, and the patient reported the development of pain in the right clavicle. She was transitioned to 3.375 g of piperacillin–tazobac-

tam, administered every 6 hours, until hospital day 3, when the creatinine level decreased to 0.59 mg per deciliter. The dose was then increased to 4.5 g administered every 6 hours. Magnetic resonance imaging (MRI) confirmed intraosseous gas in the midright clavicle (Fig. 4).

It is likely that the gram-negative rods will be identified as *F. necrophorum*, which, owing to its production of hydrogen sulfide gas, is a known cause of emphysematous osteomyelitis. Narrowing the patient's antibiotic regimen to piperacillin–tazobactam is appropriate. Although hematogenous osteomyelitis can often be treated with antibiotics alone, emphysematous osteomyelitis sometimes requires surgical débridement.

Surgical débridement was pursued to ensure adequate control of the source of infection. The results of aerobic, anaerobic, and fungal cultures of specimens obtained during surgery were negative, but material obtained on a bone swab grew *Cutibacterium acnes* (formerly known as *Propionibacterium acnes*). The patient's pain abated postoperatively. Blood cultures obtained before her transfer to our facility grew *F. necrophorum*, which is susceptible to penicillin, ampicillin–sulbactam, metronidazole, clindamycin, and imipenem. She was discharged on hospital day 9. A 6-week course of metronidazole was prescribed in order to avoid intravenous therapy; *C. acnes* was felt to be a contaminant. Ten days later, pain developed on the right side of her neck along with a purulent discharge from the site of the incision. Metronidazole was continued and vancomycin and ceftriaxone were started, but the patient had a hypersensitivity reaction to the vancomycin. Repeat incision and drainage revealed a thick gray aspirate and normal-appearing cortical bone. Cultures of samples obtained from repeat bone swabs and from the sternoclavicular joint were negative.

Although *C. acnes* is a gram-positive anaerobic bacillus that is found in normal skin flora, in this patient, given the purulent drainage, it should be treated as a pathogen. *C. acnes* is susceptible to penicillin, but cephalosporins are an acceptable alternative when long-term administration is needed.

A 6-week course of intravenous daptomycin and ceftriaxone was prescribed, and the metronida-

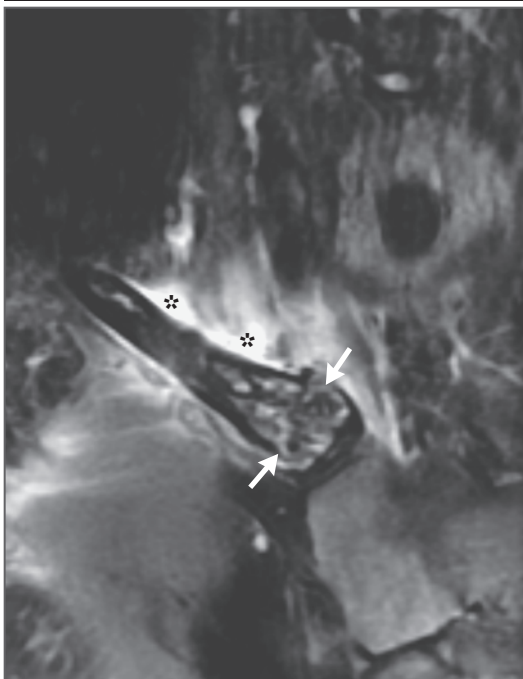


Figure 4. MRI of the Clavicle.

Multiple areas of gas can be seen, along with mild edema surrounding the bone marrow (arrows). There is also marked edema in the soft tissue (asterisks).

zole was continued. This treatment was followed by an additional 6-week course of oral amoxicillin–clavulanate. One month after the completion of her antibiotic regimen, the patient continued to have clavicular pain, but she had otherwise recovered completely.

COMMENTARY

One of the challenges for providers involved in treating a patient who presents with common symptoms is that we may be slow to recognize the point at which the common becomes uncommon. This patient's adenopathy and pharyngitis were presumed to be associated with mononucleosis, but the results of serologic tests were not consistent with acute mononucleosis. She presented several times with progressively worsening symptoms before her providers pursued additional diagnostic evaluation, which resulted in delayed diagnosis. In the interim, she underwent substantial clinical deterioration, and secondary complications developed.

The diagnosis of Lemierre's syndrome requires a high degree of clinical suspicion. The

syndrome is initially characterized by 4 to 7 days of worsening pharyngitis and should be considered in patients with persistent fever and pain on one side of the neck. A palpable neck cord can indicate thrombosis of the internal jugular vein, but this finding is neither sensitive nor specific. Ultrasonography is often used to diagnose this condition, but its sensitivity in patients with Lemierre's syndrome remains uncertain. CT and MRI, performed with the use of venous-phase contrast material, remain the standard diagnostic tests.¹

In a systematic review of 84 patients in case reports and case series,² investigators found that infection of the pharynx and tonsils was the most common source of Lemierre's syndrome, occurring in 67% of patients, with infection of the lung (occurring in 25%), larynx, ears, mastoid process, sinuses, teeth, and eyes being less common. Two more recent reviews revealed similar percentages.^{1,3} Following epithelial invasion, bacteria are believed to track along vascular planes, eventually invading the carotid sheath; subsequent inflammation is thought to trigger a local hypercoagulable process, promoting thrombosis.^{4,6} Complications of Lemierre's syndrome include sepsis, thromboembolism, pharyngitis, and infection of the central nervous system; the frequency of complications varies across studies.¹⁻³

F. necrophorum is a member of the oral microbiome in 10 to 48% of patients aged 15 to 45 years old⁷ and is responsible for 30 to 58% of published cases of Lemierre's syndrome,^{1,3} although these percentages may be overestimates due to publication bias and the known association of Lemierre's syndrome with *F. necrophorum*. In addition to causing Lemierre's syndrome, *F. necrophorum* and related species can also cause pharyngitis, odontologic infection, bacterial meningitis, cerebral abscess, and descending mediastinitis.⁶ As gram-negative bacilli of the Bacteroidaceae family, these bacteria are resistant to vancomycin but susceptible to the carbapenems and to piperacillin–tazobactam.^{6,8} Metronidazole resistance is rare but has been reported.⁹

Management of Lemierre's syndrome centers on antibiotics and supportive care. Empirical treatment should consist of a beta-lactamase-resistant beta-lactam or a carbapenem; however, the presence of hemodynamic instability should prompt the administration of a broad regimen

that includes coverage of MRSA.¹⁰ Before antibiotics were available, ligation of the internal jugular vein was a common management strategy, and the syndrome was frequently fatal. The use of broad-spectrum antibiotics is associated with low mortality (less than 2% in most recent reports).^{1,2} The benefits and risks of systemic anticoagulation have not been investigated in clinical trials, and the findings reported in retrospective studies have been inconsistent with respect to whether its use is associated with improved outcomes.^{3,7-12}

Emphysematous osteomyelitis is a rare entity that is associated with a substantial risk of ad-

verse outcomes. In a review of 25 patients, 18 of whom had serious coexisting conditions, the majority underwent surgical intervention, and 8 (32%) died.¹³

This case underscores the importance of an early index of suspicion regarding “typical” cases that evolve atypically. Had the presumptive diagnosis been revisited when the diagnostic information first diverged from what would have been expected, it seems likely that surgical intervention and chronic pain could have been avoided.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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